

One-Round Data Engineer Assignment

Production Case: March Data Accuracy, Reconciliation & Trusted Reporting

Candidate Mission

You are joining as the Data Engineer responsible for solving Arre Voice's existing March reporting mismatch. This is not a theory exercise. Submit an actionable solution that the team can use to debug, reconcile, and fix the reporting layer.

Field	Expectation
Role	Data Engineer - Product Data
Experience level	2-3 years
Preferred background	D2C / consumer app / social / content / product analytics
Time expected	4-6 hours
Submission	One PDF / Google Doc + optional SQL file
What we expect	A practical solution pack: diagnosis, SQL, final table design, RCA, data quality checks, and leadership recommendation

Hiring Expectation

In one round, the candidate should show whether they can solve the current Arre Voice data mismatch, not merely explain data engineering concepts.

1. Business Context

Arre Voice is a D2C social audio platform where users listen to audio posts, engage with creators, participate in Voicepools, and move through a product journey from acquisition to retention.

Backend and frontend teams are already capturing events. The problem is that GA4, Mixpanel, BigQuery, DynamoDB, backend logs, frontend analytics, and the data lake are showing different numbers. The company needs a trusted reporting layer that product, business, and leadership teams can rely on.

2. Current Systems

Source	Current purpose
GA4	App opens, screen views, frontend analytics events
Mixpanel	Funnels, retention, product behavior, user journey analytics
BigQuery / GBQ	Internal event storage and reporting
DynamoDB	Backend-confirmed user, content, and event records
Backend logs	Server-side confirmations, failures, retries, stream/session logs
Frontend analytics	Client-side user actions and product interaction events
Data lake	Raw and historical product lifecycle data

3. March Data Problem

Source	March 1-31 event count
GA4	10,229,448
Mixpanel	5,644,723
BigQuery / GBQ	5,000,000
DynamoDB	85,000

Known fact	Impact on analysis
GA4 and Mixpanel include combined events across Pods and Voicepools	Content-type split is currently incomplete. Candidate must design how to split going forward.
Mixpanel started tracking only from March 11	March 1-10 cannot be compared directly with Mixpanel.
Mixpanel is available only for app version 8.2.x	App-version filtering is mandatory before fair comparison.
Fair GA4 vs Mixpanel filter is March 11-31 and app version 8.2.4	Only this filtered comparison is meaningful for GA4 vs Mixpanel.

Filtered Reality

After applying fair comparison filters: GA4 = 4,234,174 and Mixpanel = 5,644,723. Mixpanel is higher by 1,410,549 events, approximately 33.3%.

4. Core Objective

The candidate must submit an actionable solution that answers these seven questions:

- Why are the numbers different?
- Which numbers are comparable and which are not?
- Which source should be trusted for each metric?
- What SQL or pseudo-SQL should be used to validate the mismatch?
- What should be the final BigQuery reporting table?
- What data quality checks should run daily?

- Can leadership trust March numbers or not?

5. Required Solution Pack

The candidate must submit all sections below. Each section must be practical, specific, and directly tied to the March problem.

A. First-Day Investigation Plan

Submit a 10-step debugging plan. Each step must include: what to check, why it matters, expected output, and the decision it enables.

Must include checks for
Date range mismatch, timezone mismatch, app version mismatch
Event name mismatch and raw vs normalized event mapping
Pods vs Voicepools split and missing content_type
iOS vs Android split and app release differences
Duplicate event firing and backend retry duplicates
Missing events and late-arriving events
Backend-generated/system events vs real user actions
Raw BigQuery tables vs transformed/reporting tables
Staging/dev data leaking into production
GA4 vs Mixpanel daily and hourly trend checks

B. Comparable vs Non-Comparable Analysis

Classify each comparison as Valid, Invalid, or Valid only after filtering. Explain the required filters and reason.

Comparison	Candidate must decide
GA March total vs Mixpanel March total	Valid / Invalid / Filtered + reason
GA Mar 11-31 v8.2.4 vs Mixpanel Mar 11-31 v8.2.4	Valid / Invalid / Filtered + reason
DynamoDB 85K vs GA 10.2M	Valid / Invalid / Filtered + reason
BigQuery 5M vs Mixpanel 5.6M	Valid / Invalid / Filtered + reason
Backend logs vs DynamoDB	Valid / Invalid / Filtered + reason
Frontend play events vs backend stream events	Valid / Invalid / Filtered + reason

C. Metric-Level Source of Truth

Define the primary source, secondary validation source, and reason for each metric.

Metric group	Metrics to cover
Acquisition and app usage	App open, screen view, campaign/referral source
Signup and user creation	Signup completed, OTP verified, user created
Audio consumption	Pod play started, Voicepool play started, valid play, listen time, play completed
Engagement	Like, comment, share, follow
Retention and attrition	D1 retention, D7 retention, dormant user, one-play exit
Leadership reporting	Final DAU/WAU/MAU, final event counts, final play/listen metrics

D. GA4 vs Mixpanel Mismatch Diagnosis

For March 11-31 and app version 8.2.4, Mixpanel is 33.3% higher than GA4. Candidate must provide top causes, verification method, and fix.

Possible cause candidate must investigate	Required response
Duplicate Mixpanel firing	How to verify and how to fix
GA4 and Mixpanel tracking different event lists	Event inclusion/exclusion comparison
Timezone or date boundary mismatch	Timezone normalization logic
Platform coverage mismatch	iOS/Android split check
Late-arriving events	Ingestion timestamp vs event timestamp check
Mixpanel counting extra lifecycle events	Event category breakdown
GA4 filtering or thresholding issue	Raw export vs UI comparison
SDK implementation difference	Release/version/code path verification
Anonymous vs logged-in identity mismatch	Identity stitching check

E. Final BigQuery Reporting Table

Design the final table that product and leadership should use: `product_analytics.fact_reconciled_events`. The design must explain how duplicate events are flagged, invalid events are excluded, frontend/backend events are matched, and `final_metric_source` is selected.

Minimum columns expected
event_date, event_timestamp, ingestion_timestamp, source, event_name, raw_event_name
user_id, anonymous_id, device_id, session_id, play_session_id
content_id, content_type, platform, app_version, environment
is_backend_confirmed, is_duplicate, is_valid_event
reconciliation_status: matched / missing_backend / missing_frontend / duplicate / invalid
final_metric_source: GA4 / Mixpanel / DynamoDB / backend_logs / BigQuery

F. SQL / Pseudo-SQL Tasks

Candidate must write SQL or pseudo-SQL for at least four of the seven tasks below. Logic matters more than exact syntax.

Task	Expected output
GA4 vs Mixpanel daily comparison	event_date, ga_events, mixpanel_events, difference, percentage_difference
Duplicate event detection	event_date, event_name, duplicate_count
Signup completed but no first play	user_id, signup_date, platform, app_version, acquisition_source
One-play exit users	user_id, first_play_date, content_id, content_type, returned_in_7_days
Frontend play vs backend stream validation	frontend_plays, backend_streams, matched_plays, missing_backend_count
Listen time from heartbeat events	listen_seconds, completion_percentage per play_session_id
App version drop-off	signup_users, first_play_users, conversion_rate by app_version

G. Daily Data Quality Checks

Create a daily checklist with logic, severity, owner, and action for these checks.

Required checks
Event volume dropped >30%; Mixpanel > GA4 by >20%; duplicate rate >5%
Backend-confirmed plays missing; app version missing; staging/dev data in production
Events delayed >24 hours; unknown event names; Pod/Voicepool content_type missing
Timezone/date boundary mismatch; backend retries increased; identity stitching failure

H. Incident Simulation: March 18 Spike

Incident

On March 18, Mixpanel events suddenly increased by 70%, but GA4 and DynamoDB stayed flat. Candidate must write a real debugging response and final RCA.

RCA field	Candidate answer required
Incident date	March 18
Impacted source	Which source and why
Impacted event(s)	Event names causing spike
Root cause	Code, SDK, duplicate firing, tracking config, or data pipeline issue
Business impact	Which dashboard/metric became unreliable
Data correction required	Backfill, exclusion, dedupe, or metric correction
Product/engineering fix	Exact owner/action
Prevention check added	Daily alert or validation rule

I. Final Leadership Recommendation

Candidate must write a note under 200 words answering: Can leadership trust March numbers? The note must state which numbers are safe, which are unsafe, mandatory filters, metrics requiring backend validation, the final BigQuery table to trust, and fixes needed before next month reporting.

6. Final Submission Format

Section	Expected output
Investigation plan	10-step practical debugging plan
Comparable vs non-comparable analysis	Table with valid/invalid/filtering decisions
Source-of-truth matrix	Metric-wise source decision
GA4 vs Mixpanel diagnosis	Cause -> verification -> fix
Final BigQuery table design	Schema and reporting logic
SQL / pseudo-SQL	Minimum 4 queries
Daily data quality checks	Logic, severity, owner, action
Incident RCA	March 18 Mixpanel spike investigation
Leadership note	Recommendation under 200 words

7. Evaluation Criteria

Evaluation area	Weight
Practical debugging ability	15%
Ability to identify valid vs invalid comparisons	15%
Source-of-truth judgment	15%
SQL / pseudo-SQL quality	20%
Reconciliation and reporting table design	15%
Data quality and monitoring thinking	10%
Leadership communication clarity	10%

8. Final Assignment Prompt

Prompt

You are the Data Engineer responsible for solving the March data mismatch at Arre Voice. GA4, Mixpanel, BigQuery, DynamoDB, backend logs, frontend analytics, and data lake numbers do not match. Diagnose the mismatch, identify valid and invalid comparisons, define source of truth by metric, write reconciliation SQL/pseudo-SQL, design the final BigQuery reporting table, create daily data quality checks, investigate the March 18 Mixpanel spike, and write a short leadership recommendation on whether March numbers can be trusted.

Important

A strong answer must be actionable. It should tell Arre Voice what to check first, which data to trust now, what SQL to run, what table to build, what needs correction, and what safeguards should run daily before leadership reporting.